Microeconomics

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Contents

1 Fundamental Concepts

Economics is the social science that studies the production, distribution, and consumption of goods and services. At its core, economics addresses the fundamental problem of **scarcity**: while resources such as labor, capital, and natural resources are limited, human wants are essentially unlimited. This creates the central challenge that all economies must navigate—how to allocate scarce resources among competing uses.

Microeconomics is the branch of economics that studies how individuals, firms, and markets make decisions about the allocation of scarce resources. Unlike macroeconomics, which examines the economy as a whole, microeconomics focuses on the behavior of individual economic agents and the interactions between them.

Core Concept 1.1: Factors of Production

The production of goods and services requires four essential factors of production:

(a) **Land** encompasses all natural inputs used in production, including agricultural land, mineral deposits, water resources, and the physical space upon which production facilities are constructed.

(b) **Labor** represents the human effort devoted to production, encompassing both physical and mental work. Labor includes the services of workers across all sectors, from agricultural laborers harvesting crops to knowledge workers developing software.

(c) **Capital** refers to goods that are produced and then used to produce other goods and services. Unlike consumer goods that satisfy wants directly, capital goods facilitate future production. Examples include manufacturing facilities, machinery, transportation equipment, and tools.

(d) **Entrepreneurship** represents the creative effort of individuals who combine the other three factors of production to create goods and services. Entrepreneurs bear risk, organize production, and drive innovation.

Core Concept 1.2: Economic Systems

An **economic system** is the mechanism by which a society decides what to produce, how to produce it, and who receives what is produced. The two most prevalent economic systems in the modern world are market economies and command economies.

(a) In a **market economy**, individuals make decisions about production and consumption, and individuals keep what they produce. This system is characterized by private property rights and prices determined by supply and demand, a framework often associated with **capitalism**.

The market economy grows through the **invisible hand** whereby individuals pursuing their own self-interest inadvertently promote the general welfare. However, market economies face challenges, including income inequality and the potential for suboptimal business practices when left unregulated.

(b) In a **command economy**, the government makes all decisions about production and resource allocation, with no private property rights, thus often associated with **communism**.

This system faces significant challenges, including lack of incentive for individuals to invest, improve quality, or innovate.

In practice, most contemporary economies, including major global economic powers such as the United States and China, operate as **mixed economies** that combine elements of both market and command systems. The relative balance between market forces and government intervention varies considerably across nations and remains a subject of ongoing economic and political debate.

Core Concept 1.3: Opportunity Cost

When making economic decisions, individuals and societies must choose among mutually exclusive alternatives. The **opportunity cost** of a choice represents the value of the best alternative forgone. This concept is fundamental to rational decision-making because choosing one option necessarily means sacrificing another.

Core Concept 1.4: Law of Increasing Opportunity Cost

The **law of increasing opportunity cost** states that as more and more of a good is produced, the opportunity cost of producing additional units increases. This occurs because resources are not equally suited for producing all goods. The resources best suited for producing a particular good are used first, and as production expands, less suitable resources must be employed, requiring increasingly larger quantities of resources to produce each additional unit.

Core Concept 1.5: Production Possibilities Curve

The **Production Possibilities Curve (PPC)** illustrates the maximum combinations of two goods that an economy can produce given its available resources and technology.

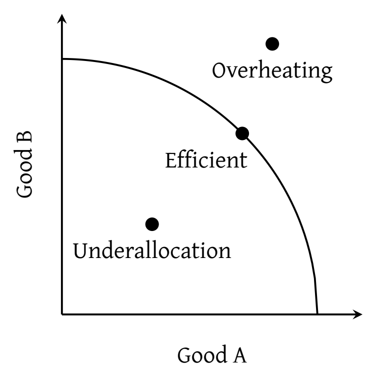

Figure 1: Production possibilities curve

(a) Points **on the curve** represent efficient production where all resources are fully utilized.

(b) Points **inside the curve** indicate underallocation of resources

(c) Points **outside the curve**—although appearing unattainable—can occur temporarily during periods of economic overheating when resources are used beyond their sustainable capacity. However, this state is not sustainable and typically leads to inflation, resource depletion, or economic crisis.

The **shape of the PPC** reveals important information about opportunity costs. A concave curve (Figure 1) indicates increasing opportunity costs, meaning that producing additional units of one good requires sacrificing increasingly larger quantities of the other good. This typically reflects the fact that resources are not equally adaptable to producing both goods.

Shifts in the PPC represent economic growth or decline. **Outward shifts indicate economic growth**, resulting from increases in available resources, improvements in technology, or enhancements in human capital. On the other hand, **inward shifts indicate economic decline**, potentially from natural disasters, war, or resource depletion.

Core Concept 1.6: Comparative and Absolute Advantage

An economy has an **absolute advantage** when it can produce more of a good than another economy using the same resources. **Comparative advantage** is when an economy can produce a good at a lower opportunity cost than another economy.

It is impossible for one economy to have a comparative advantage in both goods simultaneously. This principle, known as the **law of comparative advantage**, implies that all economies can benefit from trade by specializing in goods where they hold a comparative advantage. The gains from trade arise from this specialization and the subsequent exchange of goods.

Core Concept 1.7: Cost-Benefit Analysis

Cost-benefit analysis is a decision-making framework in which individuals and businesses compare the total benefits of an action against its total costs. The optimal decision is made when the marginal benefit equals the marginal cost.

Costs should include both **explicit costs** (*direct monetary payments*) and **implicit costs** (*opportunity costs of resources already owned*).

For instance, consider an entrepreneur who invests $200,000 to start a business and earns $255,000 in revenue during the first year. Subtracting explicit costs yields an **accounting profit** of $55,000. However, this calculation ignores opportunity costs. Suppose this individual quit a $50,000 salary job to pursue the business and foregone the 3% annual interest ($6,000) that the $200,000 could have earned in a savings account. These implicit costs total $56,000. When subtracted from the accounting profit, the **economic profit** is actually -$1,000, revealing that the entrepreneur would have been financially better off keeping the salaried position and depositing the capital in savings (*at least in the short run*).

Core Concept 1.8: Utility

Utility represents the satisfaction or pleasure derived from consuming a good or service. Utility is subjective and varies across individuals.

Marginal utility is the additional satisfaction gained from consuming one more unit of a good. For most goods, marginal utility diminishes as consumption increases—a phenomenon known as the **law of diminishing marginal utility**. For example, the first ice cream cone provides substantial satisfaction, but the tenth cone provides much less.

Core Concept 1.9: Utility Maximization Using Marginal Analysis

When faced with purchasing a combination of multiple products, a rational consumer should always incrementally purchase the product that provides the highest marginal utility per dollar spent on that marginal unit.

Core Concept 1.10: Marginal Benefit and Cost

Marginal benefit represents the additional benefit received (*often measured in monetary terms*), while **marginal cost** represents the additional cost of producing one more unit. Rational decision-making requires comparing marginal benefits and marginal costs.

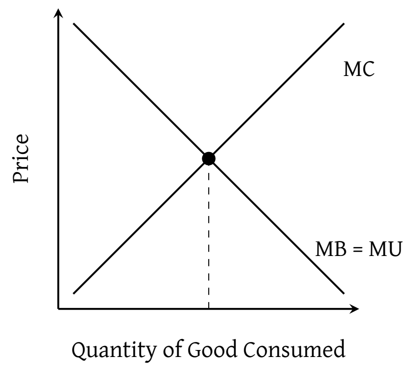

Figure 2: Marginal benefit and marginal cost

The optimal quantity occurs where marginal benefit equals marginal cost (MB = MC). At this point, the net benefit from the last unit produced equals the cost of producing it, and total net benefits are maximized.

When MB > MC, producing additional units increases total welfare. When MB < MC, producing fewer units increases total welfare.

Core Concept 1.11: Circular Flow of Income

The **Circular Flow of Income** illustrates how goods, services, and money flow between **households** and **firms** in an economy. Firms produce goods and services using factors of production provided by households and, in exchange, pay households wages, rent, interest, and profits. Households then use this income to purchase goods and services from firms, completing the circular flow.

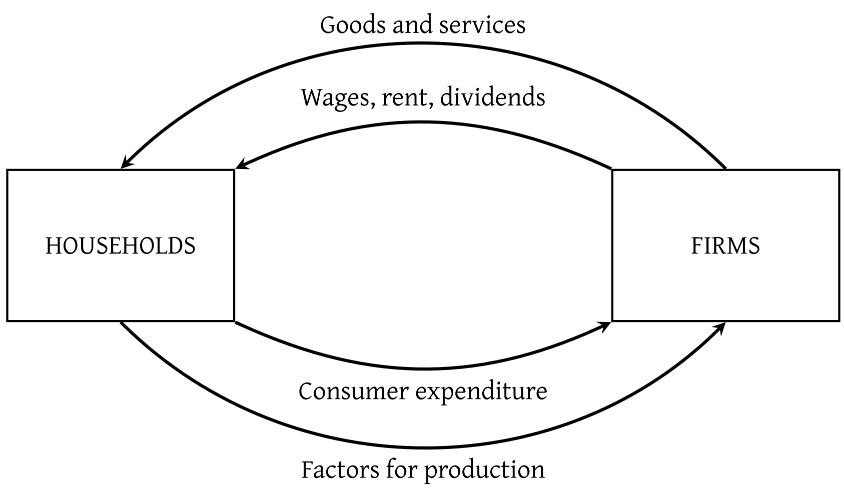

Figure 3: Circular flow of income

This model (Figure 3) illustrates the interdependence between sectors of the economy and forms the basis for measuring aggregate economic activity through Gross Domestic Product (Core Concept 2.1).

2 Supply and Demand

Core Concept 2.1: Demand

Demand represents the quantity of a good that consumers are willing and able to purchase at various prices during a given time period. The **Law of Demand** states an inverse relationship between price and quantity demanded: as prices rise, quantity demanded falls, and vice versa. This reflects both the **substitution effect** (*consumers shift to cheaper alternatives*) and the **income effect** (*higher prices reduce purchasing power*).

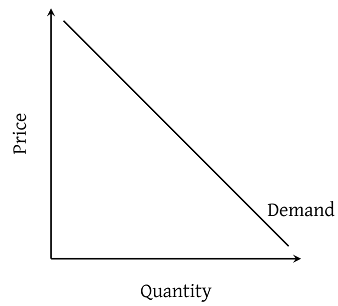

Figure 4: Demand curve

The determinants of demand, factors that can cause the entire demand curve to shift, include:

(a) **Consumer income**: Normal goods experience increased demand as income rises, while inferior goods experience decreased demand.

(b) **The number of buyers** in the market.

(c) **The prices of substitute and complementary goods**.

(d) **Consumer expectations** about future prices.

(e) **Consumer tastes and preferences**.

Core Concept 2.2: Supply

Supply represents the quantity of a good that producers are willing and able to offer for sale at various prices. The **Law of Supply** indicates a direct relationship between price and quantity supplied: higher prices incentivize producers to increase production.

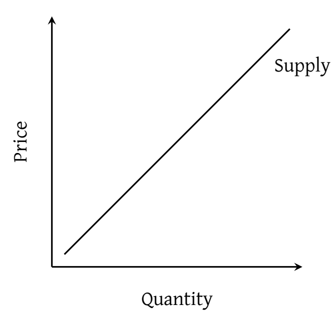

Figure 5: Supply curve

Determinants of supply include:

(a) **Production costs**, including input prices and technology.

(b) **The prices of related goods** firms could produce.

(c) **Expectations about future prices**.

(d) **The number of sellers** in the market.

(e) **Government policies**, such as taxes, subsidies, and regulations.

Core Concept 2.3: Market Equilibrium

Market Equilibrium occurs where the supply and demand curves intersect. At this point, the quantity supplied equals the quantity demanded, and the market clears. The equilibrium price and quantity are determined automatically through the price mechanism without external intervention. Correspondingly, adjustments in equilibrium price occur when supply and demand curve shifts.

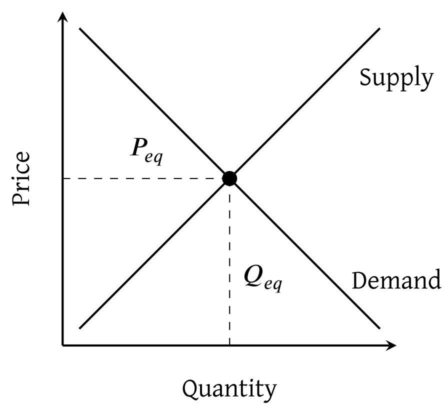

Figure 6: Market equilibrium

2.1 Elasticity

Elasticity measures how responsive one variable is to changes in another variable. In economics, elasticity quantifies the percentage change in one variable resulting from a one percent change in another variable.

Core Concept 2.4: Price Elasticity of Demand

Price Elasticity of Demand measures how much quantity demanded changes in response to changes in price and is defined by the following equation.

$$E_D = \left| \frac{\%\Delta Q_d}{\%\Delta P} \right| = \left| \frac{\frac{Q_2 - Q_1}{Q_1}}{\frac{P_2 - P_1}{P_1}} \right|$$

The following are the determinants of the price elasticity of demand.

(a) **Availability of substitutes**: More substitutes available means more elastic demand; more similar substitutes means more elastic demand.

(b) **Necessity versus luxury**: Necessities tend to have more inelastic demand.

(c) **Time frame**: Short-run demand tends to be less elastic than long-run demand.

(d) **Share of income**: Goods that represent a smaller share of income tend to have less elastic demand.

The following are the terms economists use to describe elasticity.

- **Perfectly elastic** (coefficient $= \infty$): A horizontal demand curve—price increases lead to zero demand, price decreases lead to infinite demand.

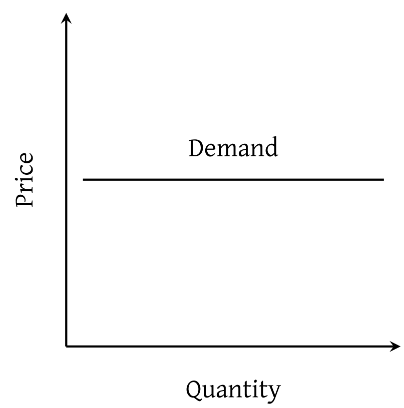

Figure 7: Perfect elasticity

- **Elastic** (coefficient > 1): Quantity demanded responds more than proportionally to price changes.
- **Unit elastic** (coefficient $= 1$): Quantity demanded responds proportionally to price changes.

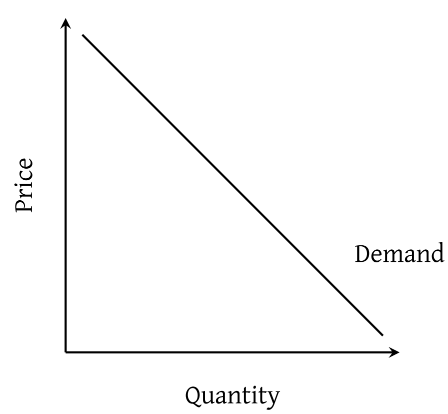

Figure 8: Unit elasticity

- **Inelastic** (coefficient < 1): Quantity demanded responds less than proportionally to price changes.
- **Perfectly inelastic** (coefficient $= 0$): A vertical demand curve—quantity demanded does not respond to price changes.

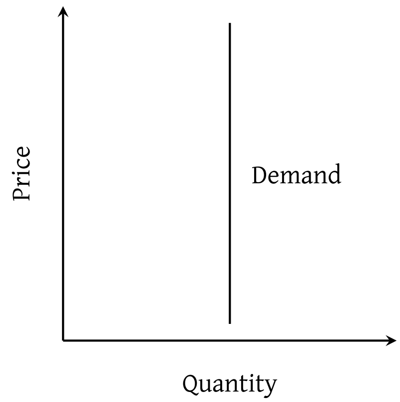

Figure 9: Perfect inelasticity

Core Concept 2.5: Total Revenue Test

The **total revenue test** is a method for determining elasticity without calculation:

- If total revenue decreases as price increases, demand is **elastic**.
- If total revenue remains constant as price changes, demand is **unit elastic**.
- If total revenue increases as price increases, demand is **inelastic**.

For a standard downward-sloping demand curve, elasticity varies along the curve: it is elastic in the upper-left portion, unit elastic at the midpoint, and inelastic in the lower-right portion.

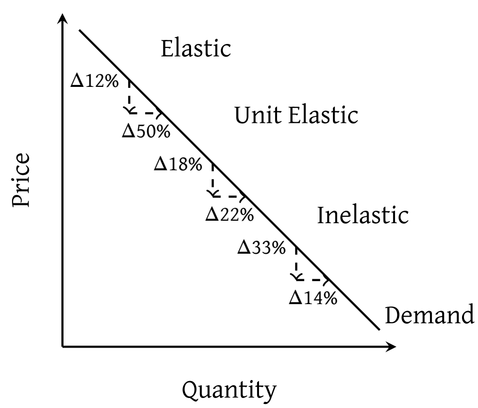

Figure 10: Total revenue test visualized

Core Concept 2.6: Price Elasticity of Supply

Price Elasticity of Supply measures how much quantity supplied changes in response to changes in price:

$$E_S = \frac{\%\Delta Q_s}{\%\Delta P} = \frac{\frac{Q_2 - Q_1}{Q_1}}{\frac{P_2 - P_1}{P_1}}$$

The primary determinant of price elasticity of supply is **time frame**. Short-run supply tends to be less elastic because producers cannot easily adjust their production capacity, while long-run supply is more elastic as firms can adjust inputs and production processes.

Core Concept 2.7: Other Elasticities

Income Elasticity of Demand measures how quantity demanded responds to changes in consumer income.

$$E_I = \frac{\%\Delta Q_{\text{demanded}}}{\%\Delta \text{Income}}$$

- If $E_I > 0$, the good is a **normal good** (*demand increases with income*).
- If $E_I < 0$, the good is an **inferior good** (*demand decreases with income*).

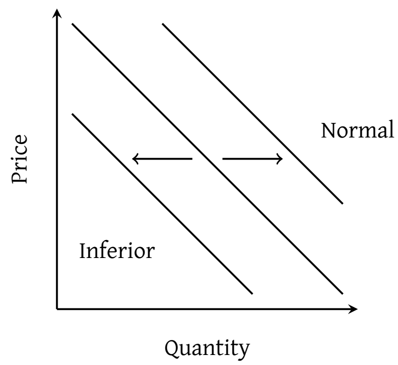

Figure 11: Demand curve movement under rising income for inferior (*left*) and normal (*right*) goods

Cross Elasticity of Demand measures how quantity demanded of one good responds to price changes in another good:

$$E_{XY} = \frac{\%\Delta Q_{\text{demanded of X}}}{\%\Delta P_Y}$$

- If $E_{XY} > 0$, X and Y are **substitutes** (*price increase in Y increases demand for X*).
- If $E_{XY} < 0$, X and Y are **complementary goods** (*price increase in Y decreases demand for X*).

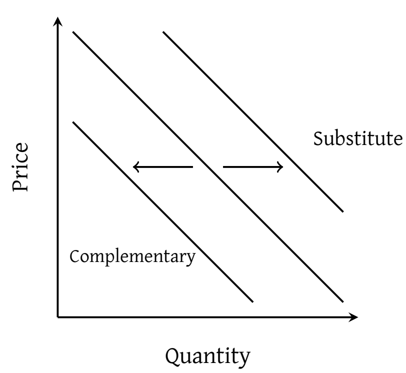

Figure 12: Demand curve movement under rising price of complementary (*left*) and substitute (*right*) goods

2.2 Surplus

Core Concept 2.8: Consumer Surplus

Consumer surplus represents the benefit consumers receive when they pay less than their maximum willingness to pay. The demand curve can be interpreted as the **marginal benefit** consumers receive from each unit consumed, reflecting the law of diminishing marginal utility.

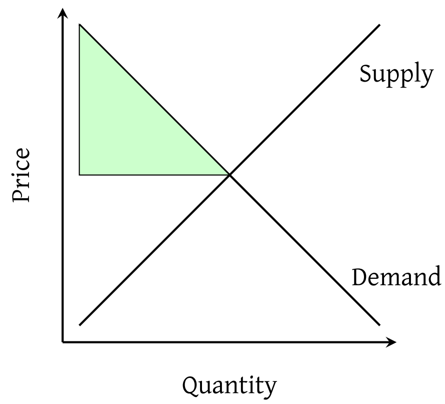

Figure 13: Consumer surplus

At market equilibrium, consumers who value the good more than the equilibrium price receive surplus because they pay the equilibrium price despite being willing to pay more. The total consumer surplus is represented by the area between the demand curve and the equilibrium price line, thus may increase or decrease depending on the demand elasticity of the good or service

Core Concept 2.9: Producer Surplus

Producer surplus represents the benefit producers receive when they sell at a price higher than their minimum acceptable price. The supply curve can be interpreted as the **marginal cost** of producing each additional unit.

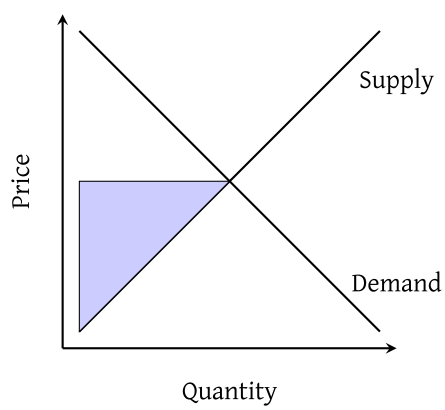

Figure 14: Producer surplus

At prices above the market equilibrium, producers receive surplus because they sell at the equilibrium price despite being willing to accept less. The total producer surplus is represented by the area between the supply curve and the equilibrium price line, thus may increase or decrease depending on the supply elasticity of the good or service.

Core Concept 2.10: Allocative Efficiency

Allocative efficiency occurs when resources are distributed in a way that maximizes the total benefit to society. In the context of consumer and producer surplus, allocative efficiency is a state in which **consumer and producer surplus is maximized**.

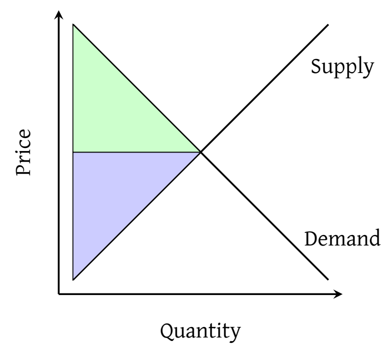

Figure 15: Allocative efficiency; consumer surplus (*green*) and producer surplus (*blue*) is maximized

Certain market interventions and imperfections can cause markets to deviate from allocative efficiency, resulting in deadweight loss and reduced social welfare.

Core Concept 2.11: Government Intervention and Surplus

Governments can intervene directly in domestic markets through policies that alter equilibrium prices and quantities.

(a) A **price ceiling** is a government-mandated maximum price below the equilibrium. When binding, a price ceiling increases consumer surplus but decreases producer surplus, creating a **deadweight loss** due to reduced quantity exchanged.

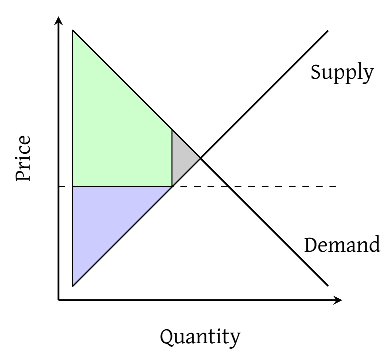

Figure 16: Price ceiling; total surplus is lowered—CS (*green*) is raised but PS (*blue*) is lowered by a greater amount—and deadweight loss (*gray*) is created

(b) A **price floor** is a government-mandated minimum price above the equilibrium. When binding, a price floor increases producer surplus but decreases consumer surplus, also creating deadweight loss.

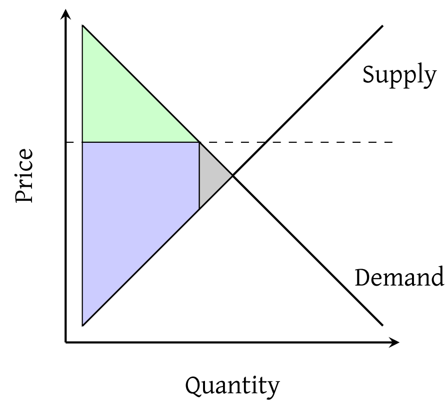

Figure 17: Price floor; total surplus is lowered—PS (*blue*) is raised but CF (*green*) is lowered by a greater amount—and deadweight loss (*gray*) is created

(c) A **per-unit tax** is a tax on each unit sold. It shifts the supply curve upward by the tax amount and consequently raises the equilibrium price, decreases both consumer and producer surplus, and creates deadweight loss. The tax revenue is captured between CS and PS. The relative incidence of the tax on consumers and producers depends on the relative elasticities of supply and demand—inelastic curves bear a greater burden.

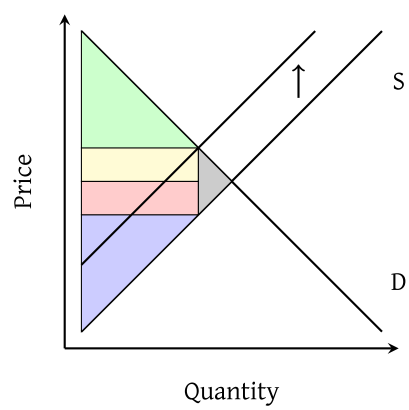

Figure 18: Per unit tax; total surplus is lowered—both PS (*blue*) and CF (*green*) is lowered and a portion is captured as tax revenue, which can be further split into the tax incidence on producers (*red*) and consumers (*yellow*)—and deadweight loss (*gray*) is created

Core Concept 2.12: International Trade and Surplus

International Trade allows countries to specialize in producing goods where they have a comparative advantage. The world supply and demand curves are determined by horizontally adding individual countries' supply and demand curves.

A country will **export** goods when its domestic supply exceeds domestic demand at the world price, and will **import** goods when domestic demand exceeds domestic supply.

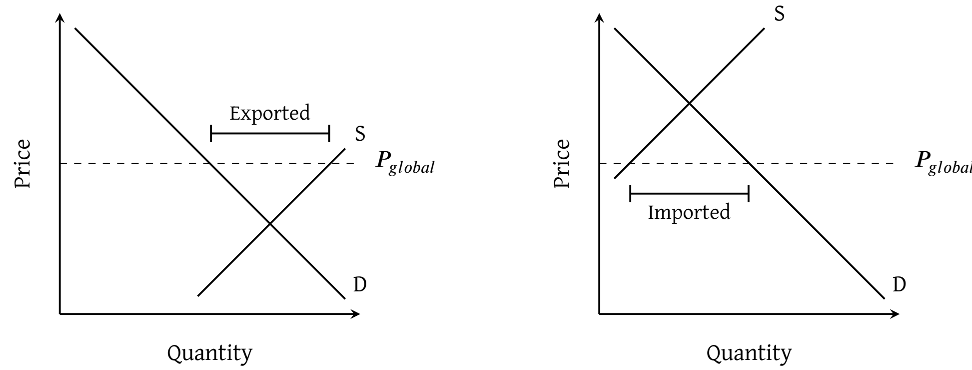

Figure 19: International trade; a country must export more goods when domestic price lies lower than global price (*left*) and import more goods when domestic price lies higher than global price

Governments can also implement public policies to regulate international trade. These interventions are typically designed to shield domestic producers from foreign competition and support emerging domestic industries.

(a) **Tariffs** are taxes on imported goods. When imposed, they increase domestic producer surplus and generate tariff revenue for the government, but decrease consumer surplus and create deadweight loss.

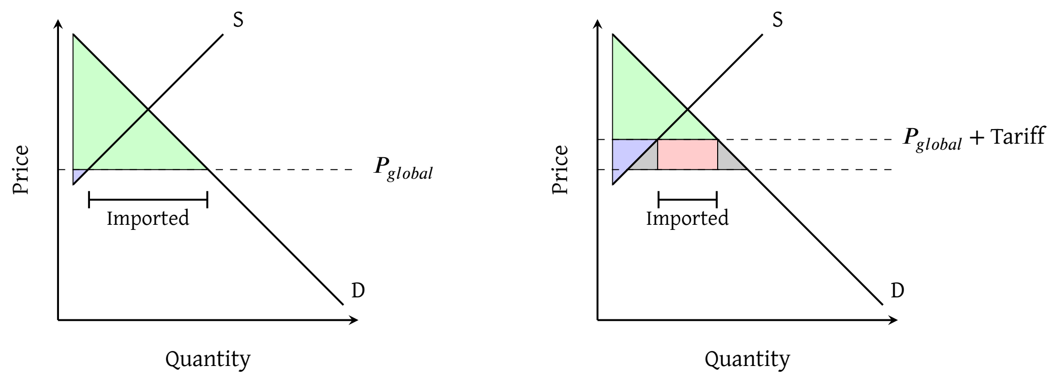

Figure 20: Tariffs; total surplus is lowered—producer surplus (*blue*) is increased, but consumer surplus (*green*) is lowered by a greater amount, a portion of which is captured as tariff revenue (*red*)—and deadweight loss (*gray*) is created

(b) **Quotas** limit the quantity of imports. They have similar effects to tariffs, except that the quota rent (*the revenue from limiting supply*) is captured by foreign exporters rather than the domestic government.

The effect of a government-enforced import quota on the domestic market is visually identical to that of a tariff (Figure 20)—both restrict the quantity of imports and drive up the domestic price above the world price. The critical distinction, however, lies in who captures the resulting revenue. Under a tariff, the domestic government collects the revenue directly as tax receipts. Under a quota, no such tax is levied; instead, the right to export is artificially scarce, so the gains accrue to foreign exporting firms in the form of higher prices—a transfer known as **quota rents**.

This means that while the deadweight loss to domestic welfare is equivalent under both policies, a quota further disadvantages the importing country by transferring what would have been government revenue abroad.

3 Production and Cost

In microeconomics, a **production function** represents the overarching relationship between **inputs** (*factors of production*) and the resulting quantity of **output**:

$$Q = f(L, K)$$

where L denotes labor (*hours worked, number of employees, etc.*) and K denotes capital (*machinery, factory space, etc.*).

3.1 Product Measures

Core Concept 3.1: Total, Marginal, and Average Product

Economists measure production output using three key metrics.

- **Total Product (TP)** refers to the total output produced.
- **Marginal Product (MP)** is the additional output gained by increasing one input by one unit.
- **Average Product (AP)** is the output per unit of input.

Here is a theoretical scenario of product as a function of input (*workers*), assuming other inputs are fixed.

Workers	Total Product	Marginal Product	Average Product
0	0	—	—
1	10	10	10
2	22	12	11
3	30	8	10
4	32	2	8
5	28	-4	5.6

Table 1: Total, marginal, and average product

When plotted against input quantity (*e.g., the number of workers*), these metrics follow a characteristic shape governed by the **law of diminishing returns**: AP and MP are predominantly downward sloping, while TP eventually plateaus and declines. Notably, MP intersects AP exactly at AP's maximum point.

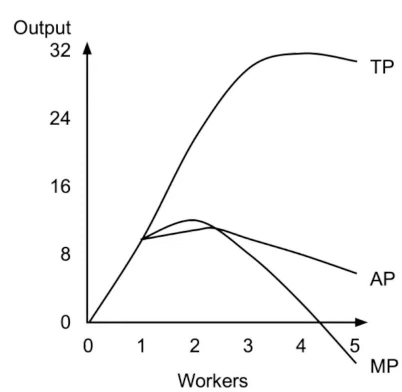

Figure 21: Total, marginal, and average product

3.2 Short Run

Core Concept 3.2: Fixed and Variable Costs

At the foundational level, economists categorize costs into the following three general categories.

- **Fixed costs (FC)** remain constant regardless of output—such as rent—and persist even at zero production.
- **Variable costs (VC)** fluctuate with output level, including payments to workers and raw materials.
- **Total cost (TC)** is simply their sum, $TC = FC + VC$.

Note that economic analysis accounts for both explicit and implicit costs.

Core Concept 3.3: Short Run Cost Curves

In the short run, the following cost curves are central to analyzing firm behavior.

- **Marginal cost (MC)** measures the additional cost of producing one more unit, defined as $MC = \Delta TC/\Delta Q$.
- **Average total cost (ATC)** is total cost per unit of output, $ATC = TC/Q$.
- **Average variable cost (AVC)** and **average fixed cost (AFC)** disaggregate into their variable and fixed components respectively, $AVC = VC/Q$ and $AFC = FC/Q$. Note that $ATC = AVC + AFC$ at every level of output.

Again, consider the following theoretical situation.

Output	TC	MC	ATC	AFC	AVC
0	$5,000	—	—	—	—
100	$10,000	$50	$100	$50	$50
200	$14,000	$40	$70	$25	$45
300	$20,000	$60	$66.67	$16.67	$50
400	$29,000	$90	$72.50	$12.50	$60
500	$45,000	$160	$90	$10	$80

Table 2: Marginal, average total, average fixed, and average variable cost

The resulting plot of MV, ATC, AVC, and AFC exhibit predictable shapes also rooted in the **law of diminishing marginal returns**.

(a) **Both ATC and AVC are U-shaped**, initially falling as output rises and fixed costs are spread across more units, then rising as diminishing returns drive up marginal cost.

(b) **AFC declines continuously**, since fixed costs are distributed over an ever-larger quantity.

(c) MC, being directly tied to marginal returns, is also U-shaped and crucially **intersects both ATC and AVC at their respective minimum points**—a consequence of the relationship between marginal and average values.

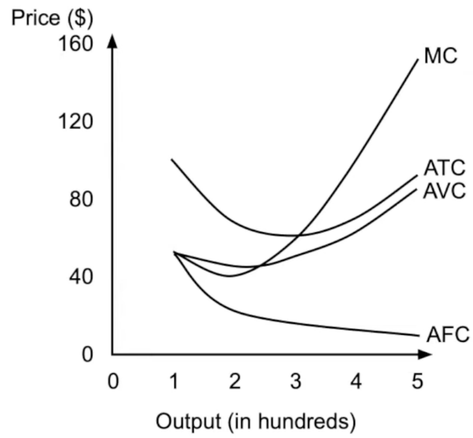

Figure 22: Marginal, average total, average fixed, and average variable cost

In perfectly competitive markets (Core Concept 4.1), a firm faces a constant market price, making marginal revenue $MR = P$. The firm maximizes profit by producing at the quantity where $MR = MC$, since any unit where $MR > MC$ adds to profit, while any unit where $MR < MC$ erodes it.

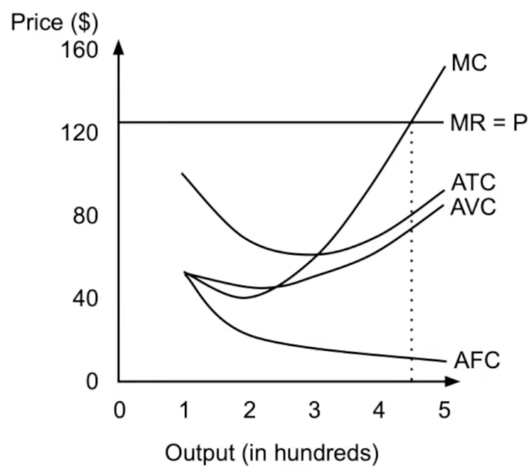

Figure 23: Profit maximization during perfect competition

3.3 Long Run

Core Concept 3.4: Long-Run Average Total Cost

In the long run, **all costs become variable** as firms are free to adjust every factor of production, including capital. The **Long-Run Average Total Cost (LRATC)** curve is constructed as the combination of many short-run average total cost (SRATC) curves, each corresponding to a different fixed level of capital (*such as factory size*)—representing the lowest attainable average cost at each output level.

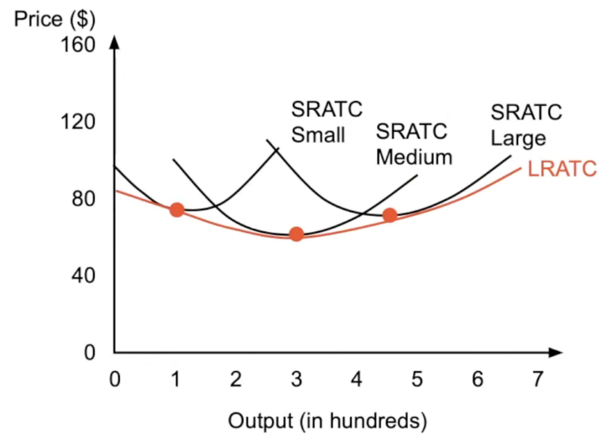

Figure 24: LRATC curve as a combination of SRATCs at differing factory sizes (*capital levels*)

As shown in Figure 24, the LRATC curve is U-shaped. This characteristic shape reflects the follow three distinct phases of production scale.

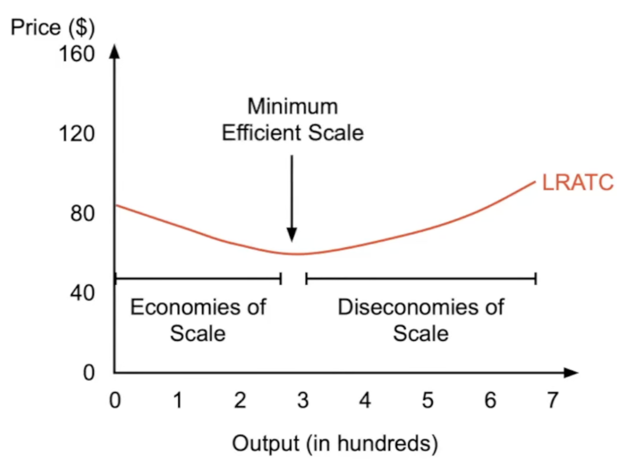

Figure 25: Economies and diseconomies of scale on the LRATC

(a) Initially, **economies of scale** cause LRATC to fall as output expands: larger operations benefit from greater worker specialization, more efficient use of capital, and bulk purchasing discounts.

(b) Beyond a certain point, **diseconomies of scale** take hold and LRATC begins to rise, driven by managerial inefficiencies and coordination difficulties that emerge as firm size grows.

(c) The output level at which LRATC reaches its minimum—the transition between these two phases—is known as the **minimum efficient scale (MES)**, and represents the most productively efficient scale of operation.

Core Concept 3.5: Returns to Scale

Returns to scale describe how output responds when *all* inputs are scaled up proportionally in the long run. Formally, suppose all inputs are multiplied by some factor $\lambda > 1$. The production function $f(L, K)$ exhibits:

$$f(\lambda L,\ \lambda K)\begin{cases} > \lambda Q & \text{increasing returns to scale} \\ = \lambda Q & \text{constant returns to scale} \\ < \lambda Q & \text{decreasing returns to scale} \end{cases}$$

Consider, for example, increasing all inputs by 70%, so $\lambda = 1.7$ and $Q' = f(1.7L,\ 1.7K)$.

- **Increasing Returns to Scale**: Output grows by a greater proportion than inputs. With $\lambda = 1.7$, output rises by more than 70%—say, 80%, so $Q' > 1.7Q$. This often arises from specialization gains and more efficient use of capital at larger scales.
- **Constant Returns to Scale**: Output grows in exact proportion to inputs. With $\lambda = 1.7$, output also rises by exactly 70%, so $Q' = 1.7Q$. This implies that doubling all inputs exactly doubles output: $f(2L,\ 2K) = 2Q$.
- **Decreasing Returns to Scale**: Output grows by a smaller proportion than inputs. With $\lambda = 1.7$, output rises by less than 70%—say, 60%, so $Q' < 1.7Q$. This typically reflects coordination difficulties and resource constraints that emerge as scale increases.

Note that while returns to scale tends to be correlated with economies of scale, they do not exactly pertain to the same situation. Economies of scale can exist even without increasing returns to scale—for example, if input costs fall as you buy in bulk (purchasing power), average cost can fall even with constant returns to scale. ***Returns to scale is purely about the physical input and output relationship, ignoring prices entirely.***

4 Perfect Competition

Core Concept 4.1: Characteristics of Perfect Competition

Perfect competition is a market structure characterized by the following assumptions.

(a) **Large number of firms**: No single firm controls a significant market share. If one producer decides to double or half their production, the total supply barely shifts.

(b) **Price-taking behavior**: Each firm sells at the market price determined by industry supply and demand equilibrium (Core Concept 2.3).

(c) **No barriers to entry or exit**: Firms can freely and easily enter or leave the market due to no patents, high start-up costs, or government regulations blocking new players.

(d) **Identical products**: All firms sell the same product, meaning there is **no brand loyalty**, no product differentiation, and products across firms are **perfect substitutes** for each other.

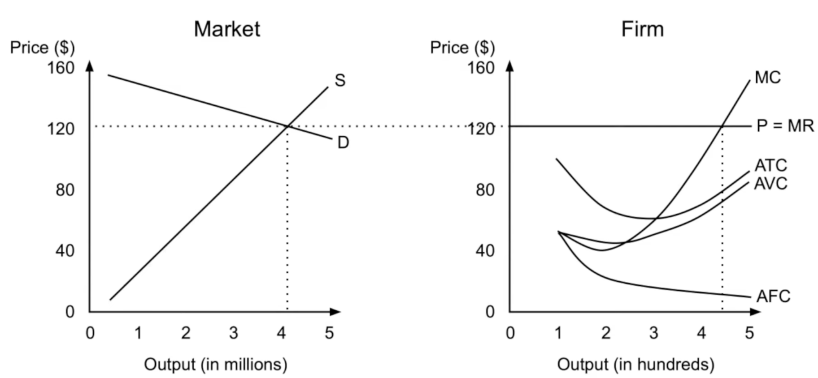

Figure 26: Market supply and demand equilibrium (*left*) setting price for individual firmMR curve (*right*); *notice the difference output scale between the market and firm*

As shown in Figure 26, the individual firm's marginal revenue curve is **perfectly elastic** (*horizontal*) at the market price, meaning that if a individual firm raises its price above market price, quantity demanded falls to zero and that the firm can sell any quantity at the market price.

Core Concept 4.2: Firm Entry and Exit

The decision for any individual firm to enter or exit a perfectly competitive market depends on profitability categorized into the following scenarios.

- If **P > ATC**, individual firms earn **economic profit**, attracting new firms to enter over time until price is pushed down due to the influx of market supply from more producers.

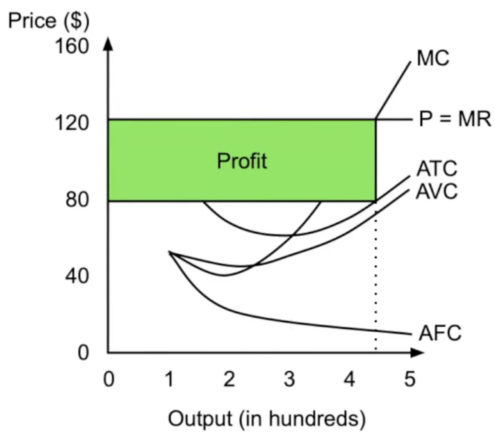

Figure 27: Price is greater than average total cost

- If **P = $\mathbf{ATC_{min}}$**, firms earn **normal profit** (*zero economic profit*), which is the equilibrium state where firms neither enter nor leave. *Mind that accounting profit would be still be positive in this scenario.*

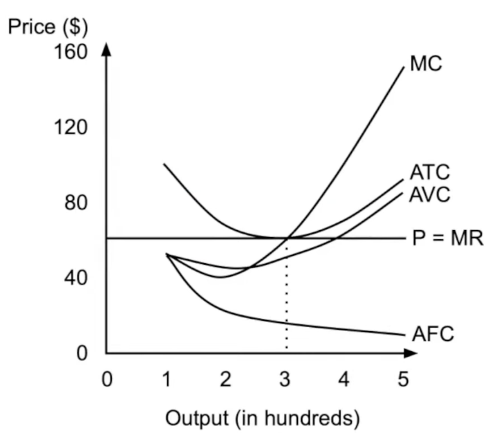

Figure 28: Price is equal to minimum average total cost

- If $\mathbf{AVC_{min}}$ **< P <** $\mathbf{ATC_{min}}$ firms **incur losses** due to high initial fixed cost but **continue producing in the short run** because they still make an economic profit with every unit in the context of AVC, which helps to cover a portion of the fixed costs, but not all.

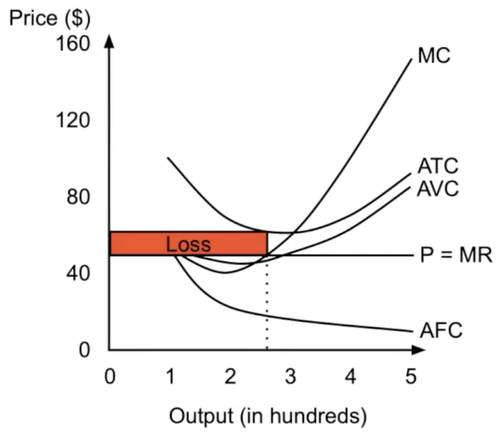

Figure 29: Price lies above minimum average variable cost but below average total cost

- If **P < AVC$_{min}$**, firms shut down immediately as the producer incurs loss with every unit of production.

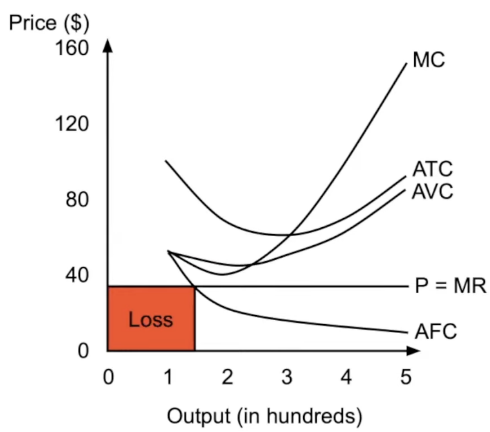

Figure 30: Price lies below minimum average variable cost; *the loss area shown only refers to the fixed cost portion which should be the only loss for an economically rational producer because any production would simply lead to greater total loss*

In the long-run, a perfectly competitive market price would converge to where **P = ATC$_{min}$**, leading to all individuals firms operating at **normal profit** or **zero economic profit**. This equilibrium state is said to be **productively efficient** because goods are produced at the minimum ATC.

If the market is not at equilibrium within the short run, new firms will enter and leave accordingly to price levels to minimum average total cost as discussed in the above scenarios.

5 Imperfect Competition

Core Concept 5.1: Imperfect Competition

Imperfect competition describes a market structure where individual firms possess a non-negligible market share, granting them a degree of **price-setting power**. Unlike the price-taking firms of perfect competition, these entities face downward-sloping demand curves; consequently, they must lower their prices to increase the volume of output sold. The following are the primary subcategories of imperfect competition.

(a) **Monopoly**: A single firm selling a product with no close substitutes. High barriers to entry allow the firm to be the sole price setter.

(b) **Monopolistic Competition**: **Many** firms selling **differentiated** products with **low barriers to entry**. *Examples include restaurants and clothing brands like Nike, Adidas, and Under Armour.*

(c) **Oligopoly**: A **small** number of firms selling identical or differentiated products. Characterized by **interdependence** and **high barriers to entry**. *Examples include smartphone manufacturers (Apple, Samsung).*

Barriers to entry are structural or legal obstacles that prevent new firms from entering an industry. These are primarily comprised of economies of scale, patents and legal statutes, prohibitive start-up costs, and exclusive ownership of key resources, such as critical commodities.

5.1 Monopoly

Core Concept 5.2: Monopoly Demand and Marginal Revenue

A **monopolist** faces the market demand curve, which is downward sloping. The MR curve lies below the demand curve because lowering price to sell more units applies to all units, not just the marginal unit.

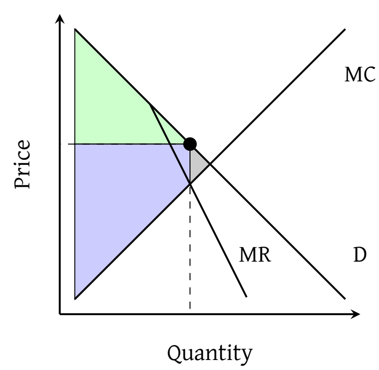

Figure 31: Monopoly supply and demand; consumer surplus (*green*), producer surplus (*blue*), deadweight loss (*gray*)

As shown in Figure 31, the monopolist produces at the quantity where MR = MC and sets price on the demand curve at that quantity, creating producer surplus at the expense of consumer surplus and generating deadweight loss.

Core Concept 5.3: Natural Monopoly

A **natural monopoly** occurs when a single firm can serve the entire market at a lower cost than multiple firms. This typically happens when LRATC curve declines throughout the relevant range of demand, meaning MC is always below average total cost.

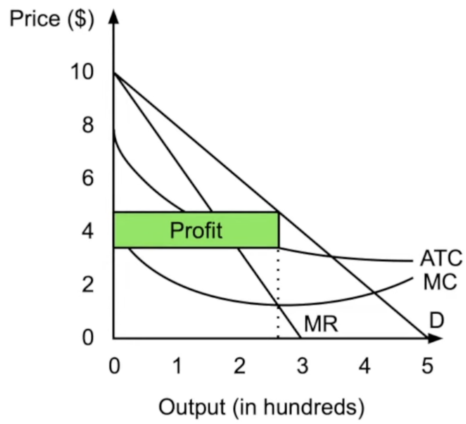

Figure 32: Natural monopoly production cost and revenue curves

Like the price setting tactics described in Core Concept 5.2, a natural monopolist will produce at the quantity where MR = MC and set price on the demand curve, generating profit as shown in Figure 32.

Core Concept 5.4: Price Discrimination

Price discrimination involves charging different customers different prices for the same product. Examples include senior/child discounts and geographic pricing (*increased price levels in urban areas compared to rural regions*).

Price discrimination increases producer surplus at the expense of consumer surplus.

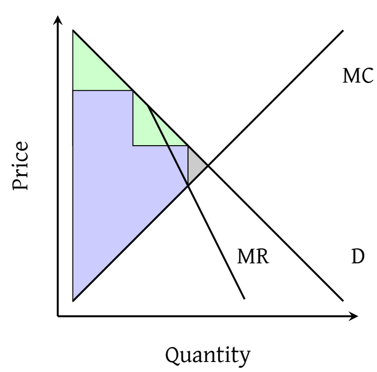

Figure 33: Price discrimination; consumer surplus (*green*), producer surplus (*blue*), deadweight loss (*gray*)

Perfect price discrimination charges each consumer their maximum willingness to pay, effectively eliminating all consumer surplus at the benefit of the producer. In this case, MR = D because the price effect (*the fact that lowering pirce to sell more units applies to all units*) becomes obsolete.

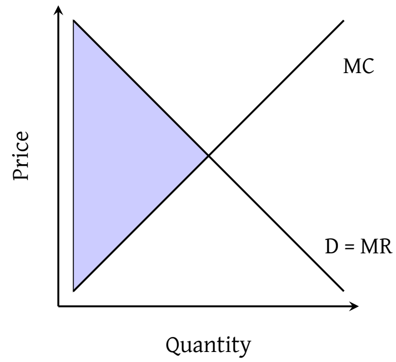

Figure 34: Perfect price discrimination; consumer surplus (*none*), producer surplus (*blue*), deadweight loss (*none*)

5.2 Monopolistic Competition

Core Concept 5.5: Monopolistic Competition

Monopolistic competition features many firms selling **differentiated** products with low barriers to entry, such as restaurants. Due to this characterisitic product differentiation, each firm or business entity faces a **relatively elastic demand curve but not a perfectly elastic demand curve**. Additionally, like monopolies, these firms produce where MR = MC and set price on the demand curve.

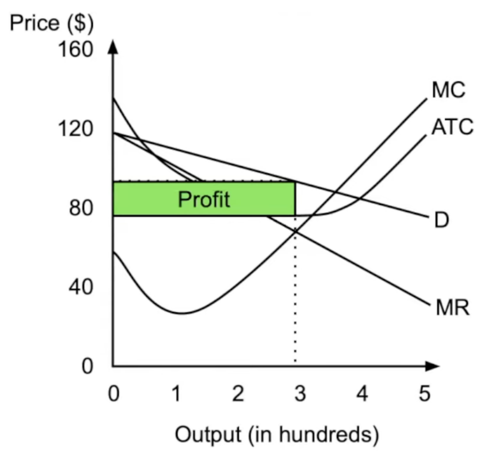

Figure 35: Monopolistic competition

As shown in Figure 35, the individual firm will operate at a profit or loss equal the distance between the ATC and D curve.

Core Concept 5.6: Long Run Equilibrium

In the long run equilibrium, new firms enter until economic profit equals zero, which is until when the demand curve is **tangent** to ATC at the quantity where MR = MC.

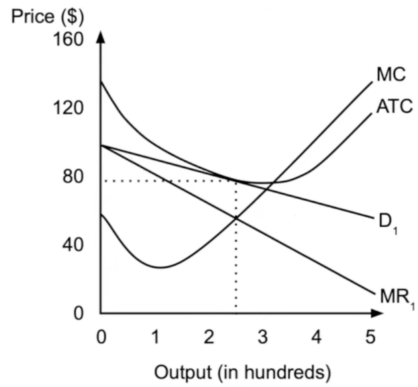

Figure 36: Long run monopolistic competition

Monopolistically competitive markets are **not productively efficient** because they produce below minimum ATC (*leaving excess capacity*) and **not allocatively efficient** because price exceeds marginal cost.

It's important to keep in mind that this long run equilibrium state is more a long run tendency rather than a state of dynamic equilibrium because, in reality, business entities may achieve a long run economic profit due to factors outside of competitors reach, such as brand loyalty or business secrets, blurring the line between monopolistic competition and monopolies.

5.3 Oligopoly

Core Concept 5.7: Oligopoly

An **oligopoly** refers to a market with a **small number of firms** with **high barriers to entry** and where firms are **interdependent**, meaning each firm's decisions significantly affect competitors.

Smartphone companies such as Apple, Google, Samsung, etc. is a nice example of oligopolies on all levels of its supply chain. Soda beverage companies such as Coca Cola and Pepsi are also good examples in the context of global logistical capabilities but becomes more of a monopolisitic competition in the context of a local beverage market.

Core Concept 5.8: Cartels

A **cartel** is a group of firms that **collude** to act like a single monopolist, restricting output and raising prices. **Antitrust laws** often restrict or prohibit cartels. However, individual firms have **incentives to cheat on agreements to increase their own profits** (*the basis of this behavior is explained in Core Concept 6.1*).

6 Game Theory: (A Brief Lesson)

Game theory analyzes strategic interactions where outcomes depend on all players' decisions.

Core Concept 6.1: Payoff Tables

Payoff tables show outcomes for each combination of **strategies**—a courses of action participants can take. The following is a payoff table of two firms choosing between high and low prices.

	Firm B Low	Firm B High
Firm A Low	(80, 80)	(130, 50)
Firm A High	(50, 130)	(100, 100)

Table 3: Firm A and Firm B payoff table; *the first and second value in each pair corresponds to Firm B and Firm A gains*

Dominant strategies yield a better outcome regardless of the choice the other participant makes. In the scenario presented in Table 3, the dominant strategy for both firms would be to **set a high price** because it provides the superior gain when looking at each choice of the other firm in isolation.

Core Concept 6.2: Nash Equilibrium

A **Nash equilibrium** occurs when neither participant can improve its outcome by changing strategy, assuming the other participant doesn't change. This results in neither player having an incentive to deviate unilaterally, which is the case presented in Table 3.

It's important to note that not all scenarios possess a Nash equilibrium.

7 Factor Markets

Factor markets study the demand and supply for the factors of production, namely labor.

Core Concept 7.1: Factor Demand

Factor demand is the quantity of a factor (*e.g., labor*) that firms are willing to purchase at each price (*e.g., wage*). Factor demand is a **derived demand**—it does not exist for its own sake but stems directly from the demand for the final good or service that the factor helps produce. *A rise in demand for housing, for instance, pulls up the demand for construction workers.* As a result, factor demand shifts in response to the following determinants.

(a) **Price of the end product:** since factor demand is derived from product demand, any change in the price of the final good ripples back to the factor market.

(b) **Productivity of the factor:** a more productive factor generates more output per unit, increasing its marginal revenue product and thus the firm's willingness to hire it at any given price.

(c) **Prices of substitute and complementary resources:** factors of production can be substitutes or complements for one another.

Namely, the general rise of and falling cost of AI and automation—a close substitute to human labor in relevant fields—shifts the demand for human labor leftwards.

Core Concept 7.2: Factor Supply

Factor supply is the quantity of a factor that owners are willing to offer at each price. It shifts in response to the following determinants.

(a) **Workforce size:** a larger population of workers eligible to supply a given factor shifts supply rightward, placing downward pressure on wages.

(b) **Education and training:** factors that require specialized skills are in limited supply by nature. As more workers acquire the necessary qualifications, supply expands and the skill premium narrows.

(c) **Alternative options available:** if returns in an alternative occupation rise, workers may reallocate toward it, reducing supply in the original market. Factor supply in any given market is always shaped by the opportunity cost of not supplying elsewhere.

(d) **Preferences for leisure:** at sufficiently high wages, workers may choose to work fewer hours rather than more—prioritizing leisure over additional income. This produces the characteristic **backward-bending labor supply curve**, where supply eventually turns leftward as the wage rises.

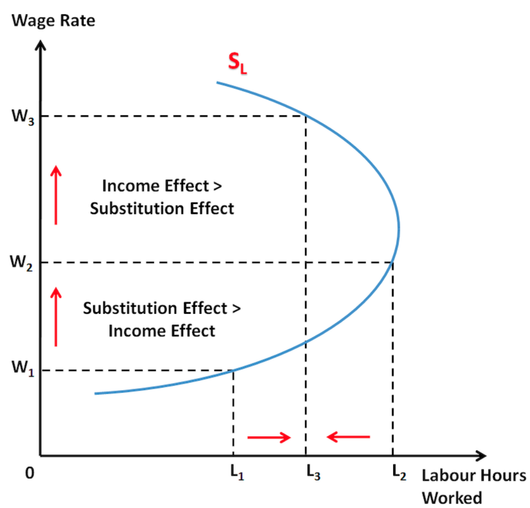

Figure 37: Backward-bending labor supply curve

Core Concept 7.3: Marginal Revenue Product and Marginal Resource Cost

Marginal Revenue Product (MRP) is the additional revenue generated by employing one more unit of a factor. In a perfectly competitive product market, MRP = MP × P because the price (*marginal revenue*) curve is horizontal.

Marginal Resource Cost (MRC) is the additional cost of employing one more unit of a factor. In a perfectly competitive **factor** market, MRC simply equals the factor price and is **constant**.

To maximize profit, firms produce and employ factors at the quantity where MRP = MRC.

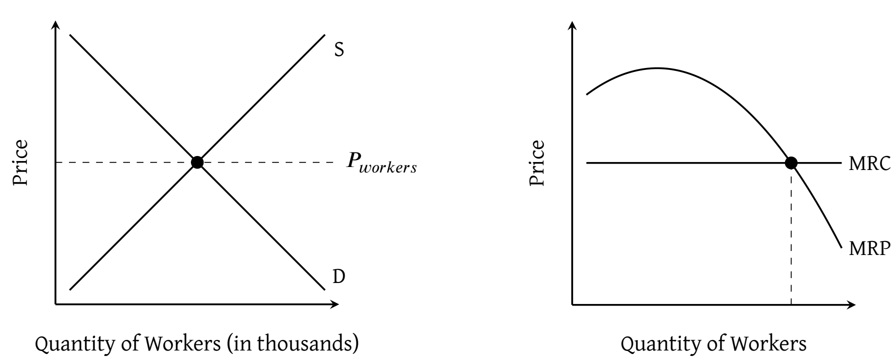

Figure 38: Perfectly competitive factor market; market (*left*) and firm (*right*)

When presented with the a choice between multiple factors, total product is maximized by allocating each successive unit of spending to whichever factor **currently** offers the **highest MP per dollar spent**.

Core Concept 7.4: Monopsony

Monopsony is the factor market analogue of a product market monopoly—just as a monopoly features a single seller, a monopsony features a **single buyer** dominating the market for a factor of production such as labor.

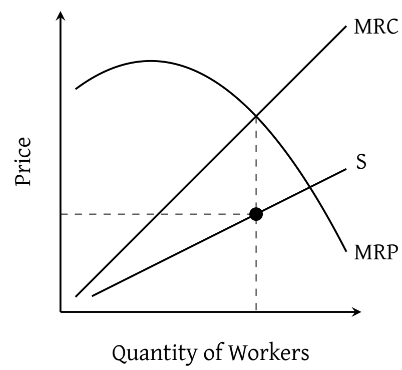

Figure 39: Monopsony plot

As shown in Figure 39, the MRC curve rises faster than the factor supply curve because hiring an additional worker requires raising the wage for all existing workers, not just the marginal hire. The profit-maximizing monopsonist hires up to the quantity where MRC = MRP, then pays the wage read off the supply curve at that quantity—a wage below the competitive equilibrium.

8 Government Intervention

Core Concept 8.1: Market Failure

Market failure occurs when market forces fail to produce the **allocatively efficient** quantity of output. This can result from externalities (*Subsection 8.1*), public goods (*Subsection 8.3*), imperfect competition (*Section 5*), or information asymmetries.

Core Concept 8.2: Government Intervention Methods

During instances of market failure, the government can intervene to restore allocative efficiency. Unlike the interventions discussed in Core Concept 2.11—where government action introduced deadweight loss into an originally efficient market—this section examines the reverse scenario: markets that begin in a state of allocative inefficiency, and the interventions used to correct it.

(a) **Lump-sum tax/subsidy**: Because it **affects only fixed costs**, it leaves marginal benefit, marginal cost, and profit-maximizing output unchanged.

(b) **Per-unit tax/subsidy**: It **shifts marginal cost or benefit**, which reduces output in the case of a tax and increases it in the case of a subsidy.

(c) **Price ceiling/floor**: When binding, it directly affects the firm's profit-maximizing output.

(d) **Regulations**: These encompass measures such as antitrust laws, which aim to promote competition within a market and eliminate monopolies.

8.1 Externalities

Core Concept 8.3: Externalities

An **externality** occurs when individuals not involved in a transaction are affected by it.

- A **positive externality** arises when a transaction generates **benefits** for third parties (*e.g. streetlights improving safety for all road users, not just those who pay for them*).
- A **negative externality** arises when a transaction imposes **costs** on third parties (*e.g. factory pollution harming the health of nearby residents*).

Core Concept 8.4: Marginal Social Cost and Benefit

Previously in Core Concept 1.10, MC and benefit curves represented only **private** costs and benefits. In reality, externalities create **social** costs and benefits.

- **Marginal Social Cost (MSC)** represents the total cost of producing one additional unit, accounting for both the producer's private costs and any costs imposed on third parties.

$$\text{MSC} = \text{Marginal Private Cost (MPC)} + \text{Marginal External Cost (MEC)}$$

Since MSC exceeds MPC in the presence of a negative externality, the market overproduces relative to the allocatively efficient level; the government can correct this by imposing a **per-unit tax** equal to the MEC, shifting MPC upward until it coincides with MSC.

In Figure 40, the deadweight loss quantifies the aggregate social cost of ***overproduction****—specifically, those units produced beyond the intersection of MB and MSC.*

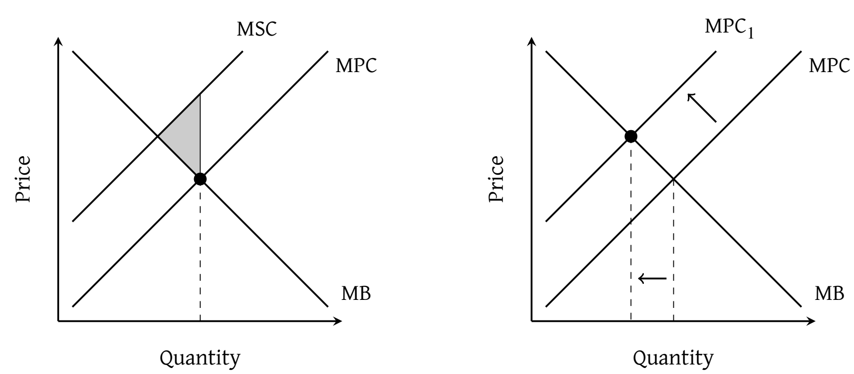

Figure 40: Marginal social cost in an allocatively inefficient market (*left*) and corresponding government intervention (*right*) via a per-unit tax

- **Marginal Social Benefit (MSB)** represents the total benefit of consuming one additional unit, accounting for both the consumer's private benefit and any benefits conferred on third parties.

$$\text{MSB} = \text{Marginal Private Benefit (MPB)} + \text{Marginal External Benefit (MEC)}$$

Since MSB exceeds MPB in the presence of a positive externality, the market underproduces relative to the allocatively efficient level; the government can correct this by providing a **per-unit subsidy** equal to the MEB, shifting MPB upward until it coincides with MSB.

In Figure 41, the deadweight loss quantifies the aggregate social benefit lost due to ***underproduction****—specifically, those potential units not produced between the market equilibrium and the intersection of MSB and MC.*

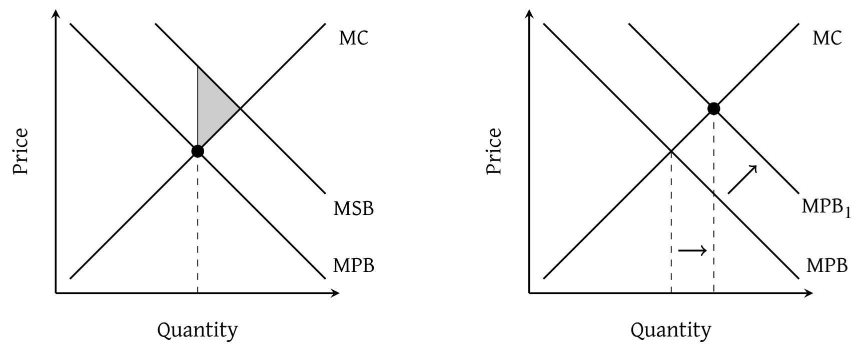

Figure 41: Marginal social benefit in an allocatively inefficient market (*left*) and corresponding government intervention (*right*) via a per-unit subsidy

8.2 Monopoly Interventions

Core Concept 8.5: General Monopoly Intervention

Since a monopoly produces where MR = MC rather than where D = MC, it **underproduces** and charges above the allocatively efficient price. The government can address this through two main interventions.

(a) A **price ceiling** set at the level where D = MC forces the monopoly to lower its price, expanding output to the socially optimal quantity.

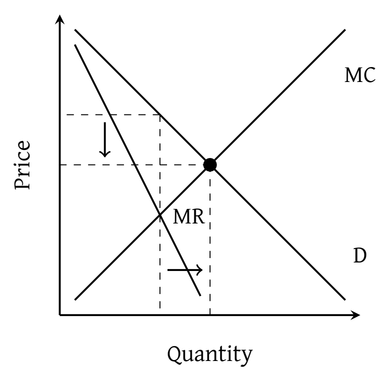

Figure 42: Government price ceiling intervention on monopolies

(b) A **per-unit subsidy** reduces the monopoly's MC, shifting it downward until MR = MC at the allocatively efficient quantity.

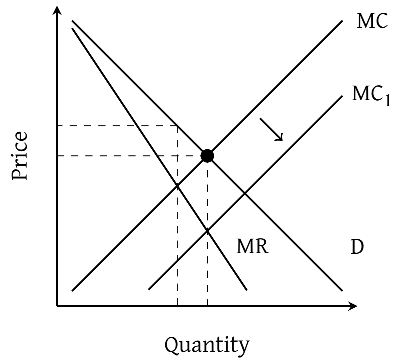

Figure 43: Government per-unit subsidy intervention on monopolies

Core Concept 8.6: Natural Monopoly Price Ceilings

Without regulation, a **natural monopoly** (*Core Concept 5.3*) produces where MR = MC, setting a price above both ATC and MC and generating an economic profit at the expense of allocative efficiency. The government can address this through two price ceiling policies, each targeting a different objective.

- **Fair return pricing**: A price ceiling set where **demand intersects ATC** compels the firm to lower its price to the point of normal profit, eliminating economic profit while keeping the firm financially viable.

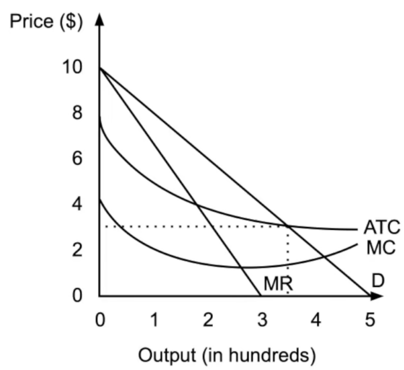

Figure 44: Fair return pricing

- **Socially optimal pricing**: A price ceiling set where **demand intersects MC** maximises total economic surplus, but drives the firm to incur an **economic loss** since price falls below ATC; the government can offset this with a **lump-sum subsidy**.

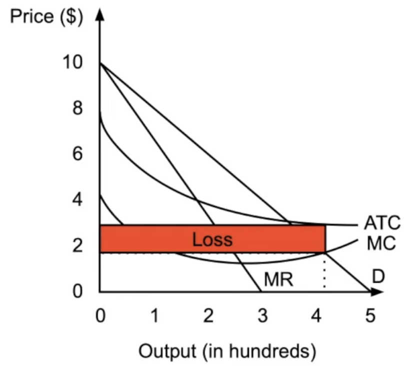

Figure 45: Socially optimal pricing

8.3 Public Goods

Core Concept 8.7: Types of Goods

Goods are classified along two dimensions: **rivalrousness** (*whether one person's consumption reduces availability for others*) and **excludability** (*whether people can be prevented from consuming the good*). Together these produce four categories:

	Rivalrous	**Non-Rivalrous**
Excludable	Private Goods (*clothing, coffee*)	Club Goods (*Netflix, gyms*)
Non-Excludable	Common Resources (*fish in the ocean*)	Public Goods (*national defense*)

Core Concept 8.8: Public Goods Issues

The **Tragedy of the Commons** arises when individuals overconsume shared **common resources** because they bear only a fraction of the total cost—overfishing is a classic example. Governments typically address this through **permits or usage regulations**.

Relatedly, the **Free Rider Problem** occurs when individuals benefit from non-excludable goods without contributing to their cost, which is why **public goods** are generally funded through **taxation** rather than voluntary payment.

8.4 Income Equality

Core Concept 8.9: Lorenz Curve and Gini Coefficient

The **Lorenz Curve** plots the cumulative share of income against the cumulative share of population.

The **Gini Coefficient (G)** summarizes the Lorenz Curve (*Figure 46*) as a single number, defined as the ratio of the area between the line of perfect equality and the Lorenz Curve to the total area under the line of perfect equality:

$$G = \frac{A}{A+B}, \qquad 0 \le G \le 1$$

where A is the area between the diagonal and the Lorenz Curve (*blue*), and B is the area under the Lorenz Curve (*blue + red*). A value of $G = 0$ indicates **perfect equality** (*all households earn identical incomes*), while $G = 1$ indicates **perfect inequality** (*a single household earns all income*). *In practice, most countries fall between 0.25 and 0.60.*

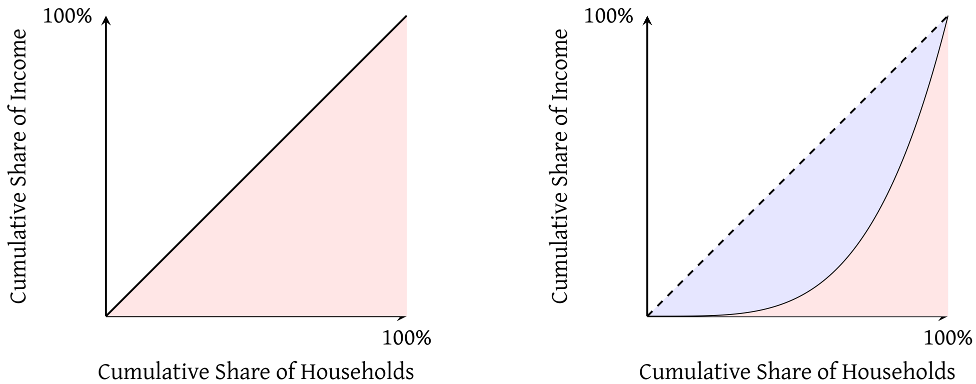

Figure 46: Lorenz curves with a gini coefficient of 0 (*left*) and >0 (*right*)

Core Concept 8.10: Government Interventions to Influence Inequality

Tax policy is one of the primary levers governments use to influence income distribution, with different structures producing markedly different effects on inequality.

- **Progressive taxes**: The tax rate **rises with income**, directly **reducing inequality** by transferring a greater burden to higher earners.
- **Proportional taxes**: A **fixed** tax rate applied to all income levels, leaving the relative income distribution **unchanged**.
- **Regressive taxes**: The effective tax rate **falls as income rises**, placing a disproportionate burden on lower earners and **worsening inequality**.

Beyond taxation, governments also intervene directly in labor markets and through spending:

- **Transfer payments**: Direct payments to lower-income households—such as welfare or unemployment benefits—raise their disposable income and reduce inequality.
- **Minimum wage and anti-discrimination laws**: Set a floor on earnings and promote equal pay, compressing the lower end of the income distribution.